
React Introduction

What Is React?

Javascript Libraries and Frameworks

TOP 5 JAVASCRIPT LIBRARIES AND FRAMEWORKS



React.js
Library



Vue.js
Framework



jQuery
Library



Angular.js
Framework



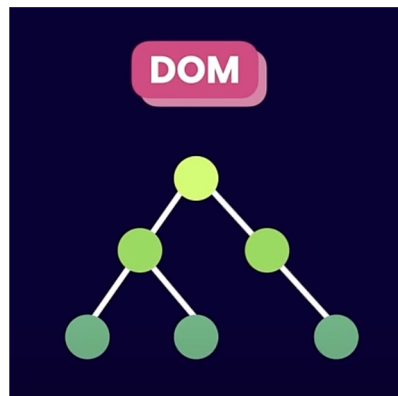
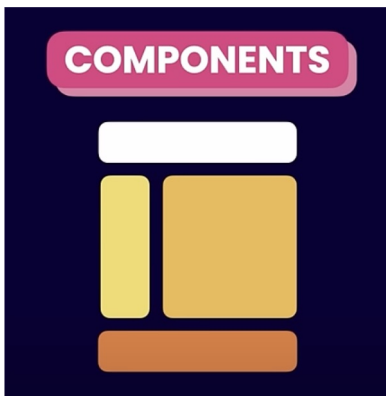
Ember.js
Framework

What is React?

- An open-source JavaScript library for building user interfaces.
- Uses reusable components to break UIs into independent, composable parts.
- Developed by Facebook
- Website: <https://react.dev/>
- Github: <https://github.com/facebook/react>

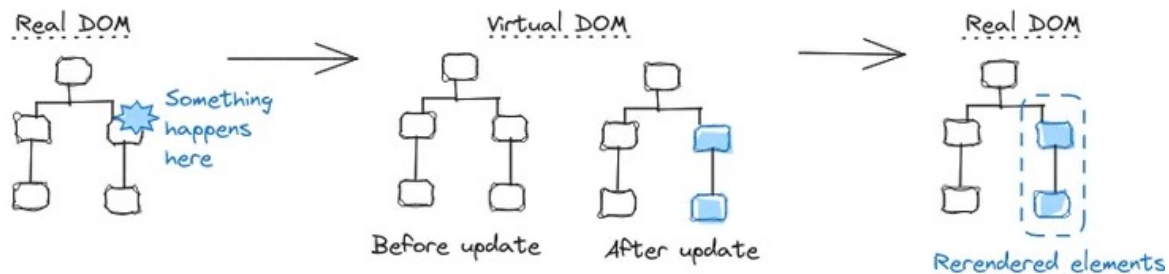
Key Features

- Virtual DOM: Efficient updates by re-rendering only what changes.
- One-Way Data Flow: Data moves from parent → child, making state management predictable and easier to debug.
- JSX: A syntax extension that lets you write HTML-like code within JavaScript.



Key Features

- Virtual DOM: Efficient updates by re-rendering only what changes.



Compute differences between the two states
of the Virtual DOM
to find the optimal way to render component
in the Real DOM

= DIFFING

Common Use Cases

- Single-Page Applications (SPAs): Gmail, Facebook, Instagram, and more.
- Mobile apps with React Native.
- Server-Side Rendering using Next.js for SEO and fast initial loads.

Setting up the Development Env

Using Node

- Need to have Node version 16 or higher
- Check the version of Node on your machine:
node --version
npm --version
- Or, download and install Node from: <https://nodejs.org/en>

Creating a React App

Two ways to create a React app

- Using the official tool provided by the React team called:
Create React APP (CRA)
- Also have another tool called: **Vite**
It's much faster and smaller bundle sizes
<https://vite.dev/guide/>

1. Create React App (CRA)

- Go to folder want to create a React app
- Then:

`npx create-react-app my-app`

Where my-app is your project. CRA will automatically:

- Create the my-app folder
- Install React, ReactDOM, Webpack, Babel...
- Create a basic file structure

- Run your react app

`cd my-app`
`npm start`

2. Vite

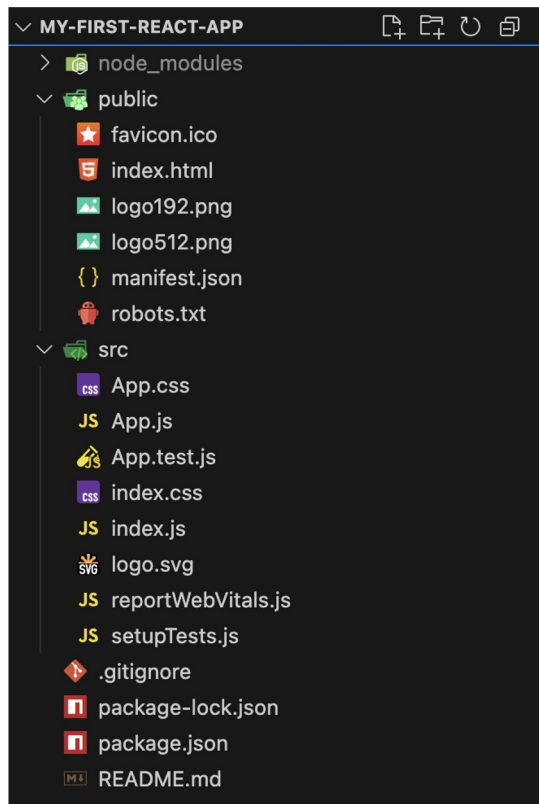
- Go to folder want to create a React app
- Interterminal:
npm create vite@latest

Then follow the prompts!

- Run your react app
cd your-react-app
npm install
npm run dev

React Project Structure

Project structure



Node modules folder

Where all the third-party libraries like react and other tools are installed.

Public folder

Where the public assets of our website exist like images, videos, etc.

Source folder

The source code of our application.

Package.json

```
package.json > ...
1  {
2    "name": "my-first-react-app",
3    "version": "0.1.0",
4    "private": true,
5    "dependencies": {
6      "@testing-library/dom": "^10.4.0",
7      "@testing-library/jest-dom": "^6.6.3",
8      "@testing-library/react": "^16.3.0",
9      "@testing-library/user-event": "^13.5.0",
10     "react": "^19.1.0",
11     "react-dom": "^19.1.0",
12     "react-scripts": "5.0.1",
13     "web-vitals": "^2.1.4"
14   },
15   > Debug
16   "scripts": {
17     "start": "react-scripts start",
18     "build": "react-scripts build",
19     "test": "react-scripts test",
20     "eject": "react-scripts eject"
21   },
22 }
```

Where we can find information about the project:

- Name of the project
- Its version
- List of dependencies
- A bunch of script

List of Dependencies

We're dependent on some libraries:

- React
- ReactDOM
- ReactScripts
- etc.

Exercises

Demo Create new React App

To create a new React app using Create React App, use the npx command, followed by create-react app, followed by a name that you want to give your new React app.

```
npx create-react-app my-new-app  
cd my-new-app
```

Summary

- Overview Front-end JavaScript frameworks and libraries
- Understand the architecture of an React application
- Scaffold out a starter React application using create-react-app, the command line tool

